



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING

COURSE CODE : 24CI3201 A.Y. 2026 - 2027

TITLE OF THE PROJECT	Invest IQ – AI-Powered Intelligent Stock Market Research and Investment Decision Support System	
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Abstract

InvestIQ – AI-Powered Adaptive Financial Research Platform is a cloud-ready web application developed to provide intelligent research and analysis of the Indian financial market, covering NSE/BSE stocks, IPOs, mutual funds, forex, and cryptocurrencies. The system integrates real-time market data, fundamental analysis, technical indicators, sentiment analysis, risk scoring, and AI-based insights to generate dynamic **BUY, HOLD, or SELL** recommendations for educational and research purposes. The project follows **Software Engineering Foundations and Agile Development** principles by adopting the **Agile SDLC**, Scrum framework, user stories, product backlog, backlog grooming, sprint planning, estimation, velocity tracking, and team communication using **Jira**. For **DevOps**, the project uses **Git workflows** for version control, **Docker** for containerization, **YAML** configuration files, and **CI/CD pipelines** for automated build and deployment, with cloud exposure through **AWS/GCP** services. **DevSecOps** and **SecOps** practices are incorporated through secure coding, **SonarQube** for static code analysis, **OWASP ZAP** for dynamic application security testing, **Trivy** for vulnerability scanning, secrets management, and security practices based on the **OWASP Top 10** and relevant compliance considerations. The AI component follows the **MLOps lifecycle**, including data ingestion, preprocessing, model training, model versioning using **DVC**, model deployment and management using **MLflow**, and continuous monitoring using **Prometheus and Grafana**, while considering AI bias, fairness, and ethical requirements. The system is designed with a **cloud-native and adaptive architecture**, allowing its data sources, AI models, analytical modules, and notification services to evolve according to changing market conditions and user requirements. Overall, InvestIQ demonstrates the practical application of **Agile Software Engineering, DevOps, CI/CD, containerization, DevSecOps, security testing, MLOps, cloud deployment, monitoring, AI ethics, and adaptive software engineering** in developing a real-world financial research platform.

Signature of the Guide

HOD